

Thiranex Data Analytics Internship Task 2

Customer Segmentation Dashboard Analysis

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1. Introduction and Objective

Understanding target audiences is key to any successful marketing and inventory strategy. Using Power BI and the Kaggle Superstore Sales dataset, this project builds an interactive Customer Segmentation Dashboard. The objective is to shift focus away from purely high-level revenue figures and drill down into customer behavior, geographic footprints, and AI-driven purchasing factors.

2. Technical Implementation and Features

- **Data Source:** Kaggle Superstore Sales dataset, focusing on regional metrics, customer fields, and purchase frequency.
- **Tooling:** Power BI, utilizing both standard visuals and advanced features like the Map visual for location mapping and the Key Influencers visual for automated machine learning insights.
- **UI/UX Approach:** Designed with a clean, soft teal theme to distinguish it from standard sales reporting and emphasize distinct customer profiles.

3. Key Findings and Behavioral Insights

- **Geographic Fragmentation:** While the store boasts a healthy base of 793 unique customers, they are highly distributed across 529 distinct cities across the United States. This indicates a broad market presence rather than hyper-local concentration.
- **Whale Customers:** The "Sales by Customer ID" chart easily spots our biggest spenders, like a customer named Sean. Finding these key people helps businesses create special VIP perks or loyalty programs to keep them coming back.
- **AI-Driven Influencers:** The smart "Key Influencers" chart shows that when people buy certain items like Tables, it triggers a massive jump in order value. In fact, it adds about \$428.90 on average to the total sales.

4. Conclusion

This dashboard provides a clear look at who is buying from the Superstore. By mapping geographic locations and letting Power BI's built-in analytics surface hidden purchase drivers, the tool helps teams replace guesswork with concrete, targeted marketing and profile-specific inventory planning.